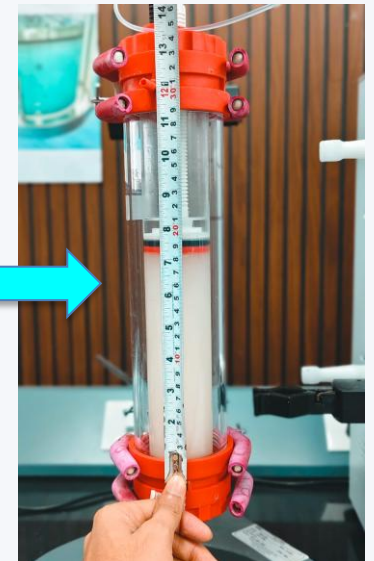


DEAE Agarose Precise
By
Protpure Tech Pvt Ltd



Column Packing

Packing factor	Value
Settled bed height	53 cm
Packed bed height	45 cm
Compression ratio	15.1 %
Packing factor	1.18
Packing velocity	350 cm/Hr
Column diameter	50 mm



Pressure Flow Characteristics

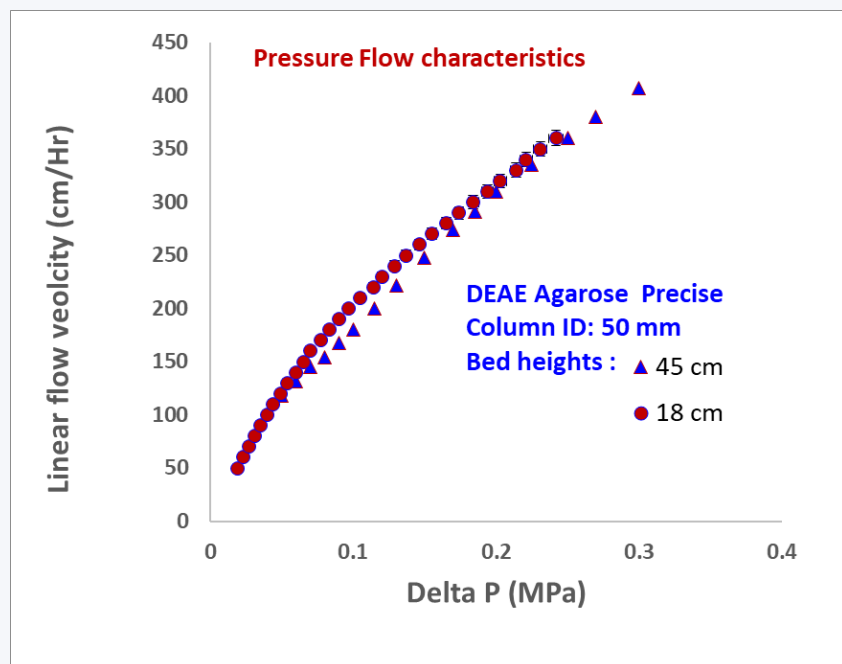


Figure: Pressure–flow characteristics of DEAE Agarose Precise packed in a 50 mm I.D. column at bed heights of 18 cm and 45 cm. The nearly overlapping pressure–flow profiles demonstrate consistent packing performance and good mechanical stability across different bed heights.

- ✓ Uniform packing quality at both bed heights.
- ✓ Good mechanical rigidity of the DEAE Agarose Precise beads.
- ✓ Predictable scale-up behavior, since pressure drop is approximately proportional to bed height.
- ✓ No evidence of bed compression or channeling over the tested flow range.

Pressure–Flow Performance

- DEAE Agarose Precise exhibits low back-pressure and excellent flow characteristics.
- Pressure–flow measurements performed in a 50 mm diameter column at bed heights of 18 cm and 45 cm demonstrated consistent hydraulic behavior, indicating robust bead structure and predictable scale-up performance. Linear velocities exceeding **350 cm/h** were achieved at pressure drops below **0.25 MPa**.

Experimental Conditions

Column Parameters

Column: 50/300 mm

Column Diameter: 5 cm

Packing Buffer

20 mM Tris 50 mM NaCl, pH 7.5

Column Evaluation

Injection: 2 mL of 1% acetone

UV monitoring at 280 nm

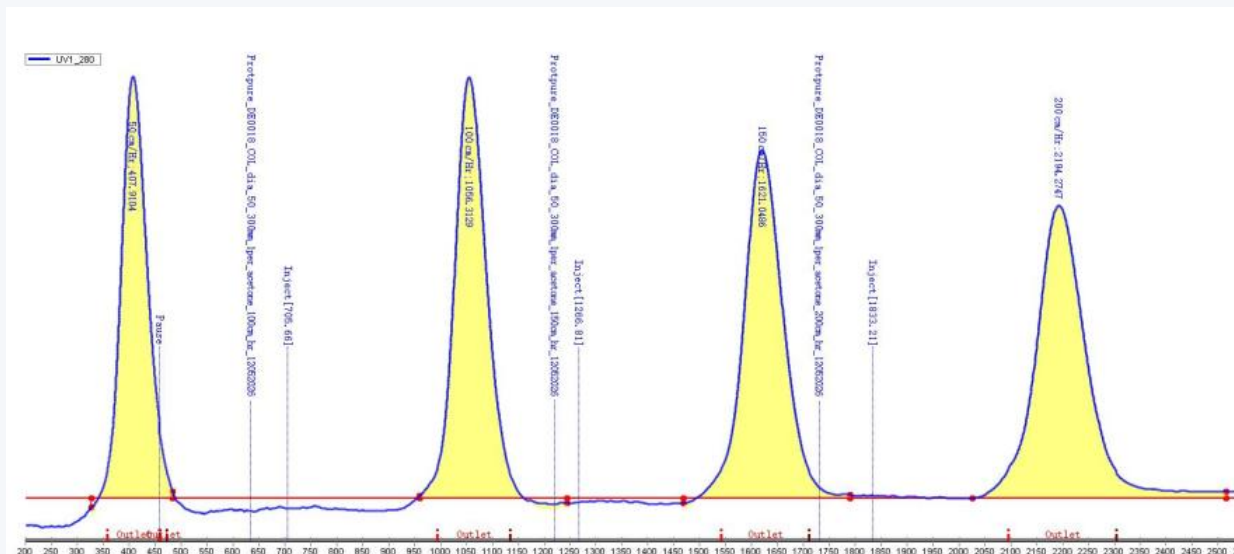


Study Objective:

“To evaluate how compressed bed height influences chromatography efficiency.”

CASE 1 — Compressed Bed Height: 18.2 cm

Column Efficiency Evaluation - Studied at linear flow velocities of 50, 100, 150 & 200 cm/hr



Integration Result:UV1_PeakTable

Number	Peak	Retention[Area mL]	Height	Plates per meter	Plate height	Asymmetry	Separation	Trailing factor
1	50 cm/Hr	407.91 456.98	7.8	1277.28	0.07829	1.3	0	0.75
2	100 cm/Hr	1056.313 554.16	7.78	7158.97	0.01397	1.19	10.83	0.78
3	150 cm/Hr	1621.049 570.02	6.44	11934.79	0.00838	1.09	7.8	0.64
4	200 cm/Hr	2194.275 618.36	5.41	14897.81	0.00671	1.16	6.5	0.76

Key Observations:

- ✓ Improved peak symmetry
- ✓ Better asymmetry values
- ✓ Higher plates per meter
- ✓ Lower plate height
- ✓ More consistent performance across flow velocities

“Improved bed consolidation resulted in better peak symmetry and higher column efficiency.”

Column Packing Efficiency

Velocity (cm/h)	Plates/m	HETP (mm)
50	1277	0.783
100	7159	0.140
150	11935	0.084
200	14898	0.067

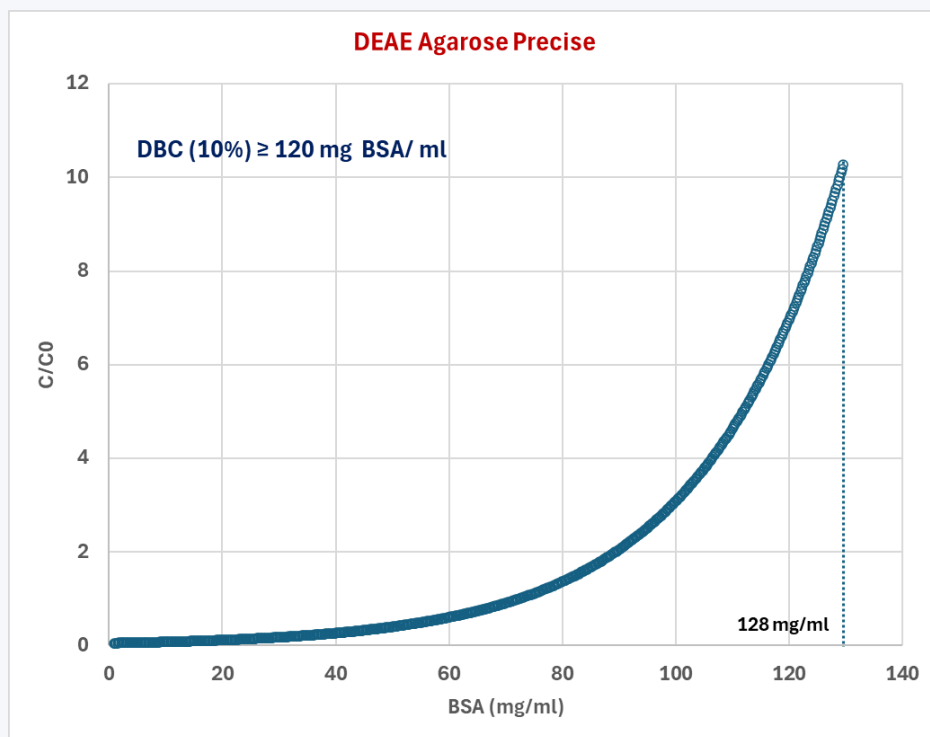
Velocity	Asymmetry
50	1.30
100	1.19
150	1.09
200	1.16

Velocity	Plate Height
50 cm/h	0.078 mm
100 cm/h	0.014 mm
150 cm/h	0.008 mm
200 cm/h	0.0067 mm

Summary

Parameter	Typical value
HETP	≤ 0.02 cm
Asymmetry	0.8–1.3

Protein Binding Capacity



Parameter	Value
Functional group	DEAE
Matrix	Cross-linked agarose
Ionic exchange capacity	14 mmol Cl ⁻ /mL
Dynamic binding capacity (10% breakthrough, BSA)	$\geq$ 120 mg/mL
Typical measured DBC (BSA)	128 mg/mL

Figure: Dynamic binding capacity (10% breakthrough) of DEAE Agarose Precise determined using bovine serum albumin (BSA) as the model protein. The resin exhibited a DBC_{10%} of approximately 128 mg BSA/mL resin under the specified operating conditions.

Technical Performance Summary

Parameter	Value	Remarks
Matrix	Cross-linked agarose	Proprietary Protpure matrix
Functional Group	DEAE (Diethylaminoethyl)	Weak anion exchanger
Particle Type	Spherical beads	Agarose-based
Ionic Exchange Capacity	TBD	To be finalized from IEC testing
Dynamic Binding Capacity (DBC10%, BSA)	128 mg/mL resin	BSA breakthrough study
Guaranteed DBC Specification	≥120 mg/mL resin	Recommended datasheet specification
Pressure Drop at 300 cm/h	~0.20 MPa	45 cm bed
Pressure Drop at 360 cm/h	~0.25 MPa	45 cm bed
Maximum Tested Linear Velocity	>400 cm/h	Stable packed bed
Bed Height Evaluated	18 cm and 45 cm	50 mm I.D. column
Compression Ratio	~15%	Representative value (53 cm → 45 cm)
Packing Factor	1.18	Representative value
Column Efficiency (N/m) @ 200 cm/h	14,898 plates/m	Acetone pulse test
HETP @ 200 cm/h	0.067 mm	Calculated from efficiency data
Asymmetry Factor @ 200 cm/h	1.16	Excellent peak symmetry
Asymmetry Range (50–200 cm/h)	1.09–1.30	Excellent packing quality
Scale-up Behavior	Excellent	Similar pressure-flow profiles at 18 and 45 cm beds
Mechanical Stability	Excellent	Demonstrated through pressure-flow and

Why DEAE Agarose Precise?

- Indigenous Indian manufacturing
- Consistent pressure-flow performance
- DBC ≥ 120 mg/mL BSA
- Scalable packing behavior
- Suitable for laboratory and pilot-scale chromatography
- Opportunity for cost-effective domestic supply